

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Artificial Intelligence and Data Science
ASSESSMENT TOOL 3 – MCQ Test

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO3 – Precision	Assessment:	Assessment Tool 3 – MCQ Test
Weightage:	30% of CIA		
CO5 Statement:	Apply N-gram models, smoothing techniques, part-of-speech tagging systems and to evaluate effective syntactic analysis. (BL-3)		
Student Name:	Y.VARUN KUMAR REDDY	Reg. No.:	192424230
Section / Batch:	AIDS SLOT-B	Date:	17-08-2026

Code of Conduct

I B.Joshna(192472097) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- Read each question carefully before selecting your answer.
- This assessment consists of 50 multiple-choice questions.
- Do not use external resources, AI tools, textbooks, or assistance from others during the assessment unless explicitly permitted by the instructor.
- Upload Score Card in GCR and submit score card printout.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
Context-Free Grammars (CFG)	CFG components, productions, terminals, non-terminals, recursion, sentence generation	1 – 10
Sentence Construction and Agreement	Parse trees, sentence structure, grammatical agreement, syntactic validation	11 – 20
Subcategorization	Verb frames, complement requirements, argument structure analysis	21 – 25
Top-Down Parsing	Parsing strategy, derivation process, recursion issues, grammar expansion	26 – 30
Earley Parsing	Predictor, Scanner, Completer operations, ambiguity handling	31 – 35
Feature Structures	Feature representation, unification, constraint checking, agreement enforcement	36 – 40
Probabilistic Context-Free Grammars (PCFG)	Rule probabilities, ambiguity resolution, probabilistic parsing	41 – 48
Integrated Syntax Analysis	CFG + Feature Structures + PCFG integration in NLP applications	49 – 50

Annexure A – MCQ Test

<https://forms.gle/WSv28zfaU5B2aWcy6>

Rubrics for Calculation

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Number of Correct Answers	100%	45 to 50 correct answers	38 to 44 correct answers	25 to 37 correct answers	0 to 24 correct answers
Accuracy	100%	Excellent (90–100%)	Good (75–89%)	Average (50–74%)	Poor (Below 50%)

No. of Questions	Number of Correct Answers	Accuracy	Total %
50			

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Test

Y. Varun
192626230

30
50

- x 1) Recursive Production rules allow hierarchical sentence $\rightarrow B$
- x 2) subject verb fragment $\rightarrow A$
- x 3) finite state Automata $\rightarrow C$
- x 4) Morphological $\rightarrow A$
- ✓ 5) left recursion causing infinite expression $\rightarrow B$
- ✓ 6) [B] User's Preset, Scanner & Compiler operation.
- x 7) Regular expression matching $\rightarrow A$
- ✓ 8) Choose the parse tree with max rule Probability [A]
- ✓ 9) Finite state Transducer [B]
- ✓ 10) CFG + feature structure + subcategory [A]
- x 11) Remove all PP [C]
- ✓ 12) Authorized employee to update [D]
- x 13) [C] The verb violates the subcategorization
- ✓ 14) [B] left recursion causing infinity recursion [B]
- ✓ 15) Dynamic stack Construction Combining Scanner & Compiler [D]
- x 16) Number feature Conflict bwn sub & obj [A]
- x 17) Test 1 because A has max Probability [A]
- ✓ 18) highly recursive grammar with unlexical Production [A]
- ✓ 19) [D] Proxy Parsing handler
- x 20) CFG Parsing - subcategory checking [C]

- ✓21) [C] Introduce a Compound noun Production
- ✓22) [A] Main subject phrase
- ✓23) [D] Verb subcategory frame
- ✓24) [A] shift reduce
- x25) [A] Predictor
- x26) [B] Maximum likely distribution
- ✓27) [A] Tree A because it has highest probability
- ✓28) [A] Context free grammar only
- ✓29) [B] recursive top down Parser
- x30) [A] Maximum Frequency matching
- ✓31) [A] S - NP/VP → on, NP - the
- x32) [D] Engineer Develops the software
- x33) [C] She gave the customer
- ✓34) [D] Expand S into NP VP
- ✓35) [A] Predictor
- ✓36) [B] uniform false because feature
- ✓37) [A] The Phase with the highest overall
- ✓38) [B] Recursive decent Parsing
- ✓39) [A] Phase A because it satisfies the Constraints
- ✓40) [B] Reject the Phase because grammatical agreement's Constraints
- x41) [B] Der NP Det/N
- x42) [A] Det
- ✓43) [C] completer
- x44) [A] sentence Accepted because Person matches
- x45) [A] The maximum Ratio of Production rules
- ✓46) [A] CFG supports recursive non phrase grammar NP → NP VP
- ✓47) [C] structured (syntactic) ambiguity
- x48) [C] Recursive grammar expansion
- ✓49) [A] The verb gave recursive additional Components
- ✓50) [A] V NP PP